

Multiple Choice

1. **Answer:** B
Options: A) -5; B) -4; C) -3; D) -2
Note: Correct option: B - -4
2. **Answer:** D
Options: A) None; B) One; C) Four; D) Five
Note: Correct option: D - Five
3. **Answer:** B
Options: A) 8; B) 120; C) 240; D) 24
Note: Correct option: B - 120
4. **Answer:** B
Options: A) 1; B) 2; C) 3; D) 4
Note: Correct option: B - 2
5. **Answer:** D
Options: A) $(S, +, \times)$ is closed under addition modulo 10.; B) $(S, +, \times)$ is closed under multiplication modulo 10.; C) $(S, +, \times)$ has an identity under addition modulo 10.; D) $(S, +, \times)$ has no identity under multiplication modulo 10.
Note: Correct option: D - $(S, +, \times)$ has no identity under multiplication modulo 10.

True / False

6. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
7. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: ' $4\mathbb{Z}, 2 + 4\mathbb{Z}$ '.
8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 0. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** $-\sqrt{34}$. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key